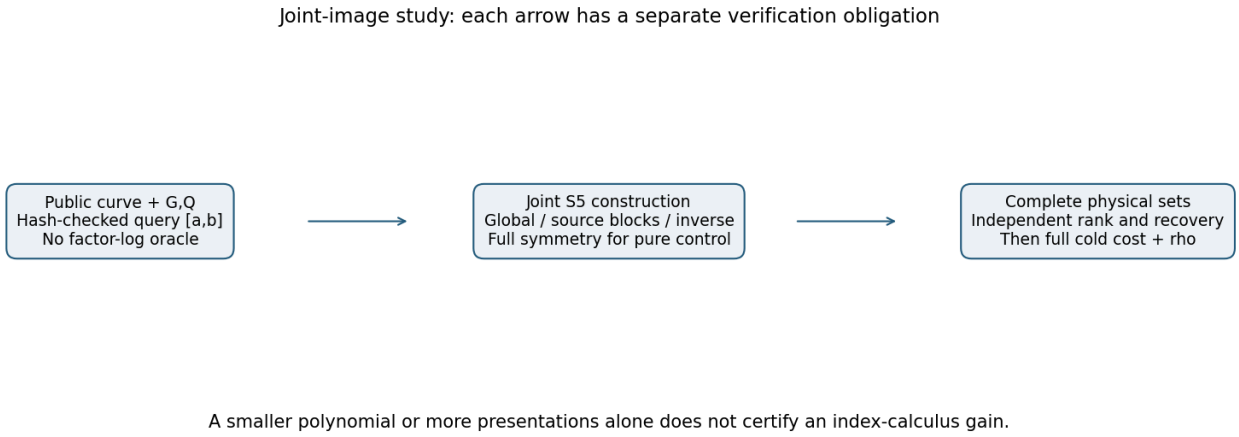
Faithful quartic discriminant proof passes

quartic-factored-lean-v6 passes its pinned Lean 4.34.1 build and transitive-axiom gate on orbit-v3/QuarticDiscriminant.lean (../../proof/orbit-v3/QuarticDiscriminant.lean). The gate audits nine declarations: seven named theorems and two local associativity/commutativity instance definitions. There are zero failing and zero trusted-native declarations. The mathematical proofs use only propext, Classical.choice and Quot.sound; the two instances use no axioms. Actual logs (../../results/proof-checks/quartic-factored-v6/) retain the source hash, build and complete axiom listing.



The proof preserves the universal commutative-ring statement: the standard quartic discriminant evaluated on four elementary symmetric coordinates equals the product of the six squared root differences. It also proves that a nonzero value implies all six pairs of roots are distinct. It does not assume an integral domain or a converse for rings with zero divisors.

The earlier direct expansion exceeded the unchanged 8 GiB allocation cap. The new proof first factors addition of a fourth root using cubic coefficients, proves the cubic discriminant and cubic evaluation identities, then combines them using small square identities and associativity / commutativity. This changes proof organization, not its conclusion or hypotheses. Failed combined/v4 builds and the pre-execution v5 PATH failure remain preserved. The successful run uses one worker, the busy reservation, 8 GiB and a 900 s watchdog.

This closes the specific polynomial factorization/nonrepetition proof gap. It does not prove finite-field membership, correspondence of the complete elimination ideal with physical relations, complete root enumeration, matrix rank, DLP recovery or a complexity/speed claim. Those bridges remain explicit in the proof map (../../proof/PROOF_MAP.md).